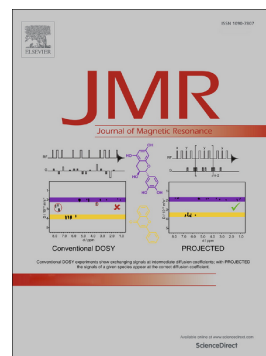


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PII: S1090-7807(26)00142-4

DOI: <https://doi.org/10.1016/j.jmr.2026.108153>

Reference: YJMRE 108153

To appear in:

Received date: 21 June 2026

Revised date: 21 July 2026

Accepted date: 7 August 2026

Please cite this article as: J. Baek, S. Min, J. Kim, et al., Efficient iridium N-heterocyclic carbene catalysts for signal amplification by reversible exchange in dimethyl sulfoxide, (2026), <https://doi.org/10.1016/j.jmr.2026.108153>

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Efficient Iridium N-Heterocyclic Carbene Catalysts for Signal Amplification by Reversible Exchange in Dimethyl Sulfoxide

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Abstract

Hyperpolarization by signal amplification by reversible exchange (SABRE) is a promising route toward fast, low-cost, and repeatable sensitivity enhancement for nuclear magnetic resonance (NMR) spectroscopy and magnetic resonance imaging (MRI). Because dimethyl sulfoxide (DMSO) is biocompatible and dissolves a broad range of organic and biological substrates, performing SABRE directly in DMSO could substantially widen the scope of parahydrogen-based hyperpolarization toward biomedical applications. To date, however, DMSO has been regarded as an inefficient SABRE medium for the conventional [Ir(IMes)(COD)Cl] catalyst, which provides only modest enhancements relative to methanol. Here we report two new iridium N-heterocyclic carbene (NHC) catalysts—the [Ir(COD)Cl] complexes of 1,3-dibenzylimidazol-2-ylidene (Ir-Imid) and 1,3-dibenzylbenzimidazol-2-ylidene (Ir-Benzimid)—that deliver the highest ¹H SABRE enhancements reported in DMSO to date. Across five representative substrates, the new catalyst improves the enhancement factor for protons at the same position by about 1.23 to 3.24 times compared to Ir-IMes. Density functional theory (DFT) calculations of the catalyst–substrate–parahydrogen complexes relate these trends to the average axial and equatorial Ir–ligand bond lengths and provide rational design guidance for further development of SABRE catalysts that operate efficiently in DMSO.

Keywords: Hyperpolarization; Signal amplification by reversible exchange (SABRE); Parahydrogen; Iridium N-heterocyclic carbene catalyst; Dimethyl sulfoxide; Density functional theory

1. Introduction

Nuclear magnetic resonance (NMR) is one of the most powerful spectroscopic techniques for elucidating the structure of organic, inorganic, and biological molecules, and for probing molecular dynamics in complex environments. The NMR signal arises from the population difference between Zeeman energy levels in an applied magnetic field. Because this population difference follows the Boltzmann distribution and scales with field strength, stronger magnets are required to achieve higher signal-to-noise ratios, which entails high costs for constructing and maintaining superconducting magnets. To overcome this intrinsic sensitivity limitation, a range of hyperpolarization techniques has been developed to generate non-equilibrium spin populations far exceeding the thermal value.^{1, 2}

Among these methods, parahydrogen-based hyperpolarization has attracted particular attention because it can be performed under ambient conditions and in real time.^{3, 4} Signal amplification by reversible exchange (SABRE) is a non-hydrogenative variant in which the latent spin order of parahydrogen is transferred to a substrate without altering its chemical identity.⁵ In SABRE, both parahydrogen and the substrate bind reversibly to an iridium center, and polarization is relayed through the scalar coupling network of the transient complex, most efficiently at low magnetic field.⁶ Since the introduction of iridium N-heterocyclic carbene (NHC) catalysts, in particular [Ir(IMes)(COD)Cl] (IMes = 1,3-bis(2,4,6-trimethylphenyl)imidazol-2-ylidene), SABRE has become a robust and widely used hyperpolarization platform,^{7, 8} and systematic ligand studies have correlated catalyst performance with the steric and electronic properties of the NHC ligand.⁹

Most SABRE studies have been carried out in methanol because of its high polarization efficiency. Methanol, however, is poorly suited to biological applications, which has motivated the search for alternative solvents such as water, ethanol, and DMSO.¹⁰⁻¹² In parallel, the substrate scope of SABRE has been expanded enormously, from simple N-heterocycles to a wide range of functional groups, neat liquids, and amino acids,¹³⁻¹⁵ including nitriles and isoquinoline derivatives of analytical interest,^{16, 17} while the SABRE-Relay strategy extends hyperpolarization to substrates that lack a suitable metal-binding site,¹⁸ enabling trace and even sub-micromolar detection by NMR.¹⁹ Beyond protons, performing the exchange at microtesla fields within a magnetic shield (SABRE-SHEATH) delivers high levels of ¹⁵N and ¹³C polarization,^{20, 21} including for biomedically relevant probes such as metronidazole and pyruvate.^{22, 23} The strong dependence of SABRE efficiency on the solvent reflects the central role of the reversible ligand-exchange kinetics that govern polarization transfer. Among the candidate media, dimethyl sulfoxide (DMSO) is especially attractive: it is biocompatible and possesses a high dissolving power that opens new possibilities for hyperpolarizing diverse substrates undergoing chemical exchange at the SABRE catalyst. DMSO has also been shown to act as a co-ligand that assists polarization transfer to heteronuclei such as ¹³C, making it an important solvent for both polarization transfer and downstream metabolic imaging.²⁴

Despite this promise, catalyst development for SABRE in DMSO remains limited. To date only [Ir(IMes)(COD)Cl] has been examined in any detail in this medium, and it provides enhancement factors much smaller than those obtained in methanol.¹² This limitation is significant because SABRE has rapidly matured into a practical analytical tool: it has enabled real-time monitoring of chemical and biological reactions and the sensitive detection of drugs and drug candidates,²⁵⁻²⁸ and recent advances in catalyst and substrate engineering have extended it to

aqueous, heterogeneous, and recyclable platforms, to multidentate chelating substrates, and even to metal-complex targets,²⁹⁻³¹ as summarized in several comprehensive accounts of the technique.³²⁻³⁴ New catalysts tailored to DMSO are therefore needed to bring these capabilities to a biocompatible medium and to extend the SABRE hyperpolarization toolbox. Here we report two new iridium NHC catalysts that operate efficiently in DMSO and substantially outperform [Ir(IMes)(COD)Cl], in some cases more than doubling the proton enhancement for specific substrates. We further rationalize the observed trends through density functional theory (DFT) calculations of the catalyst–substrate–parahydrogen complexes. These results suggest a practical strategy for designing improved SABRE catalysts in DMSO and broaden the applicability of parahydrogen-based hyperpolarization.

2. Experimental

2.1. Materials and general procedures

All reactions were carried out under a nitrogen atmosphere unless otherwise stated. Commercial reagents and solvents were used as received. ¹H and ¹³C NMR spectra used for compound characterization were recorded in CDCl₃ at 300 MHz and 75 MHz, respectively; the full spectra are provided in the Supporting Information (Figs. S1–S4). The proposed structures of Ir-Imid and Ir-Benzimid catalysts were supported by the combined ¹H and ¹³C NMR data, which were consistent with the expected proton and carbon environments of the complexes. Chemical shifts (δ) are reported in ppm, and coupling constants (J) in Hz.

2.2. Synthesis of the imidazolium ligand precursor

1,3-Dibenzyl-1H-imidazol-3-ium bromide. Imidazole and anhydrous K₂CO₃ were dissolved in acetonitrile under nitrogen and stirred for 20 min at room temperature. Benzyl bromide was added

dropwise, and stirring was continued at room temperature for 20 h. The mixture was filtered and washed with a small amount of acetonitrile, and the solvent was removed under reduced pressure to give a light-yellow oil (yield 95%). ^1H NMR (CDCl_3 , 300 MHz): δ 10.80 (s, 1H), 7.47–7.31 (m, 10H), 7.20–7.19 (d, J = 1.5 Hz, 2H), 5.55 (s, 4H).

2.3. Synthesis of the iridium catalysts

(1,3-Dibenzylimidazol-2-ylidene)(η^4 -1,5-cyclooctadiene)chloroiridium(I) (*Ir-Imid*). The imidazolium salt was dissolved in acetonitrile, and $[\text{Ir}(\text{COD})\text{Cl}]_2$ and K_2CO_3 were added. The solution was stirred at 90 °C for 4 h, then filtered and washed with a small amount of acetonitrile. The solvent was removed under reduced pressure, and the residue was purified by column chromatography (silica gel; dichloromethane/methanol = 30:1) to give the product (yield 61%). ^1H NMR (CDCl_3 , 300 MHz): δ 7.38–7.34 (m, 10H), 6.60 (s, 2H), 5.74–5.73 (m, 2H), 5.64–5.56 (m, 2H), 4.73–4.71 (m, 2H), 2.99–2.97 (m, 2H), 2.20–2.00 (m, 4H), 1.77–1.43 (m, 4H). ^{13}C NMR (CDCl_3 , 75 MHz): δ 181.17 (Ir–C), 136.51 (C_{arom}), 129.08, 128.48–128.35 (CH_{arom}), 120.73, 120.70 (CH_{imid}), 85.26, 84.68 (CH_{COD}), 54.53, 54.44 (NCH_2), 52.95, 52.14, 33.73, 33.50, 29.93, 29.67 (CH_2 , COD).

(1,3-Dibenzylbenzimidazol-2-ylidene)(η^4 -1,5-cyclooctadiene)chloroiridium(I) (*Ir-Benzimid*). The benzimidazolium salt was dissolved in acetonitrile, and $[\text{Ir}(\text{COD})\text{Cl}]_2$ and K_2CO_3 were added. The solution was stirred at 90 °C for 2 h, then filtered and washed with a small amount of acetonitrile. The solvent was removed under reduced pressure, and the residue was purified by column chromatography (silica gel; dichloromethane/methanol = 30:1) to give the product (yield 92%). ^1H NMR (CDCl_3 , 300 MHz): δ 7.41–7.30 (m, 10H), 7.03–6.95 (m, 4H), 6.19–6.03 (m, 4H), 4.83–4.76 (m, 2H), 2.95–2.93 (m, 2H), 2.18–2.14 (m, 2H), 2.06–1.97 (m, 2H), 1.79–1.71 (m, 2H), 1.52–1.39 (m, 2H). ^{13}C NMR (CDCl_3 , 75 MHz): δ 193.18 (Ir–C), 135.97, 135.04, 128.86 (C_{arom}),

127.82, 127.16, 122.61 (CH_{arom}), 111.15, 87.40, 86.75 (CH_{COD}), 53.86 (NCH_2), 53.00–52.59, 33.41, 33.17, 29.50, 29.24 (CH_2 , COD).

2.4. SABRE hyperpolarization and NMR measurements

All ^1H NMR spectra were recorded on a Bruker spectrometer operating at 300 MHz and controlled by TopSpin. For the SABRE experiments, $[\text{Ir}(\text{IMes})(\text{COD})\text{Cl}]$ or one of the two new iridium complexes (3.5 mM) was used as the catalyst precursor, together with a substrate (35 mM)—acetonitrile, pyridine, or a pyridine derivative—in DMSO-d_6 . Hyperpolarization was performed using 50% parahydrogen generated with a home-built converter. The pre-catalyst/substrate solution was first activated by bubbling parahydrogen through it for 20 min. After activation, the sample was exposed to parahydrogen for 1 min at a chosen polarization-transfer field in the range 0–120 G under 1 atm, and was then transferred to the spectrometer for detection. Enhancement (amplification) factors were obtained by comparing the hyperpolarized signal with the corresponding thermal-equilibrium signal; the complete field-dependent profiles are provided in the Supporting Information (Figs. S5–S24).

2.5. Computational details

Density functional theory (DFT) calculations were performed on the temporary catalyst–substrate–parahydrogen complexes formed by the three iridium catalysts (Ir-IMes, Ir-Imid, and Ir-Benzimid) with pyridine, nicotinic acid, 1-aminoisoquinoline, isonicotinic acid hydrazide (isoniazid), and acetonitrile. All structures were optimized with the BP86 functional using the LanL2DZ basis set in Gaussian 16. Solvent effects for DMSO and methanol were treated with the SMD continuum solvation model. For each optimized complex, the minimum energy, axial and equatorial Ir–ligand bond lengths, dipole moment, and polarizability were evaluated; full results are tabulated in the Supporting Information.

3. Results and discussion

3.1. SABRE with the reference Ir-IMes catalyst in DMSO

We first established a baseline by performing SABRE in DMSO with the well-known [Ir(IMes)(COD)Cl] catalyst (Fig. 1a). Five representative substrates—pyridine (Py), nicotinic acid (NA), 1-aminoisoquinoline (1-AIQ), isoniazid (INH), and acetonitrile (ACN)—were examined (Fig. 1b); these include N-heterocycles, a nitrile, and pharmaceutically relevant molecules previously studied by SABRE in other media.^{12, 16, 17} The proton enhancement factors were modest in all cases, remaining below about 7.5-fold. Pyridine, which typically shows the largest enhancement in methanol, gave only a roughly fivefold enhancement, and acetonitrile was enhanced by less than twofold. Among the tested substrates, isoniazid exhibited the largest enhancement, a trend that differs from that observed in other solvents, and its enhancement showed no simple, monotonic dependence on the polarization-transfer field (Fig. 1b and Figs. S5–S24). Some of these values are consistent with previous reports;¹² minor discrepancies in the absolute enhancement are attributed to differences in the NMR platform, since the 60 MHz systems used previously involve a shorter sample-transfer time between the polarization-transfer and detection stages than the 300 MHz system used here.

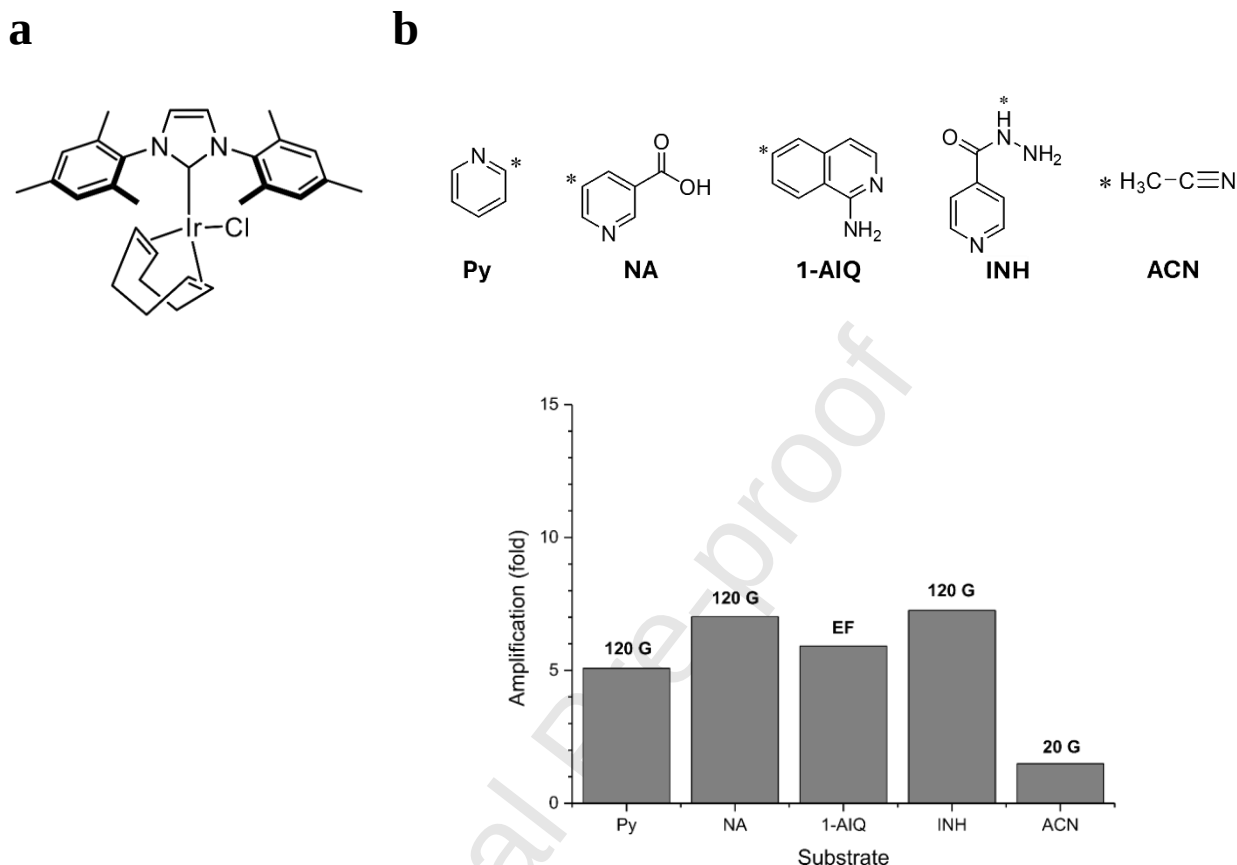


Fig. 1. (a) Structure of the reference Ir-IMes catalyst. (b) Substrates used for SABRE—pyridine (Py), nicotinic acid (NA), 1-aminoisoquinoline (1-AIQ), isoniazid (INH), and acetonitrile (ACN)—and the corresponding proton enhancement factors obtained in DMSO at the indicated polarization-transfer fields. “EF” stands for “earth’s field”, which means 0.5G. Asterisks mark the most strongly polarized proton in each structure, whose signal was used to construct the graph.

3.2. The new Ir-Imid catalyst

To obtain a catalyst with improved efficiency in DMSO, we synthesized a symmetric imidazol-2-ylidene-based iridium catalyst, Ir-Imid (Fig. 2a). Under conditions identical to those used for Ir-IMes, all five substrates were tested (Fig. 2b). At Earth’s magnetic fields (0.5 G), Ir-Imid produced enhancement factors of 8.46 fold for pyridine, 6.72 fold for nicotinic acid, 10.29 fold for 1-aminoisoquinoline, 11.50 fold for isoniazid, and 3.22 fold for acetonitrile. The proton position,

where amplification is at its maximum, showed values ranging from 1.23 (nicotinic acid) to 3.24 (pyridine) times, which are significantly higher than those achieved when using the Ir-Imes catalyst, a known efficient conventional SABRE catalyst. For comparison, additional tests with other common catalyst classes (PHOX and Crabtree-type complexes) gave no appreciable polarization for these substrates, indicating that Ir-Imid is specifically well suited to SABRE in DMSO. Notably, the largest enhancements were obtained at lower polarization-transfer fields, consistent with the level-anti-crossing description of SABRE polarization transfer,³⁵ and suggesting that the field that optimizes transfer differs among the complexes—an observation supported by the DFT analysis below.

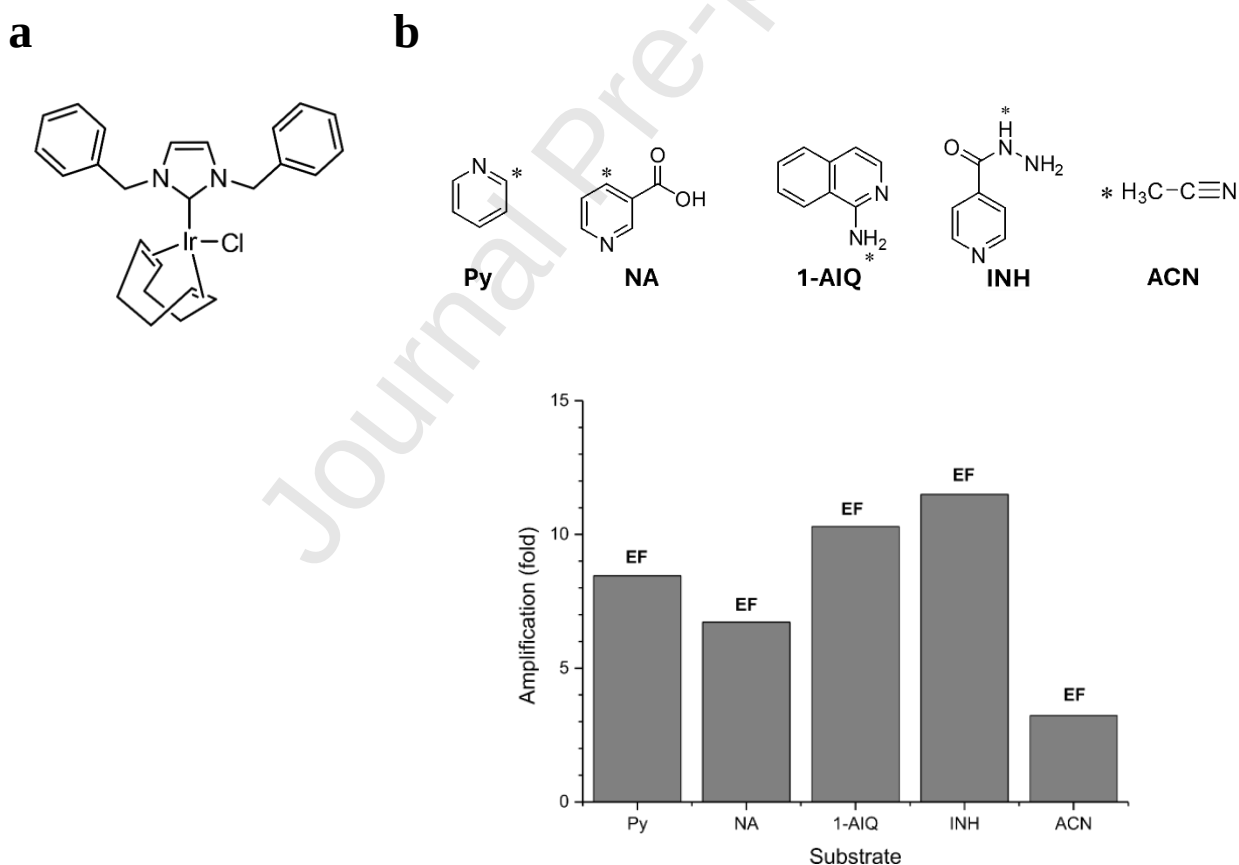
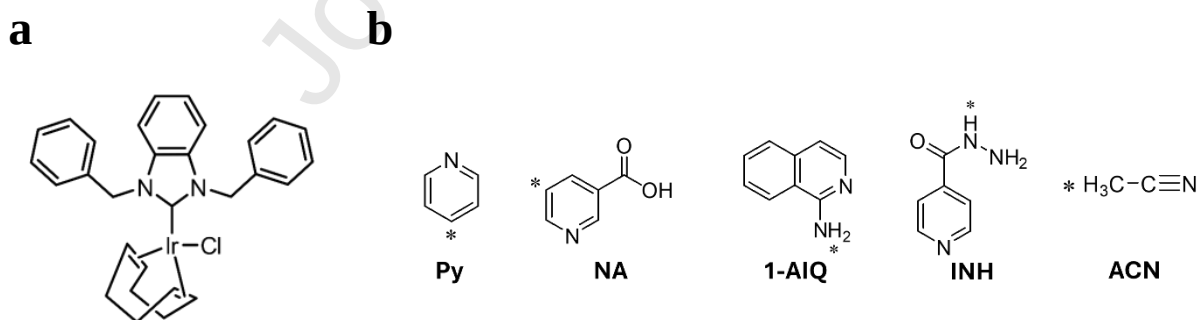


Fig. 2. (a) Structure of the new Ir-Imid catalyst. (b) Substrates used for SABRE and the corresponding proton enhancement factors obtained in DMSO at the indicated polarization-transfer fields. “EF” stands for “earth’s field”, which means 0.5G. Asterisks mark the most

strongly polarized proton in each structure, whose signal was used to construct the graph.

3.3. The new *Ir-Benzimid* catalyst

To explore the design space further, we synthesized a benzimidazol-2-ylidene analogue, Ir-Benzimid (Fig. 3a), and evaluated it with the same five substrates (Fig. 3b). Ir-Benzimid produced enhancement factors of 8.28, 12.66, and 4.76 fold for nicotinic acid, isoniazid, and acetonitrile, respectively, exceeding the corresponding values obtained with Ir-Imid. By contrast, the enhancement factors for pyridine (2.79 fold) and 1-aminoisoquinoline (8.22 fold) were lower than those observed with Ir-Imid. Thus Ir-Benzimid did not uniformly outperform Ir-Imid but exhibited selective advantages for specific substrates. This result is noteworthy because it runs counter to the conventional expectation that benzimidazol-2-ylidene ligands are less effective than imidazol-2-ylidene ligands for SABRE. As a representative example, Fig. 3c compares the thermal (black) and hyperpolarized (red) ^1H spectra of isoniazid obtained with Ir-Benzimid, showing clear signal amplification for the labeled protons. Taken together, these data underscore a central point of catalyst design for SABRE: the optimal ligand structure is solvent - and substrate-dependent, and trends established in one solvent system do not necessarily transfer to another.



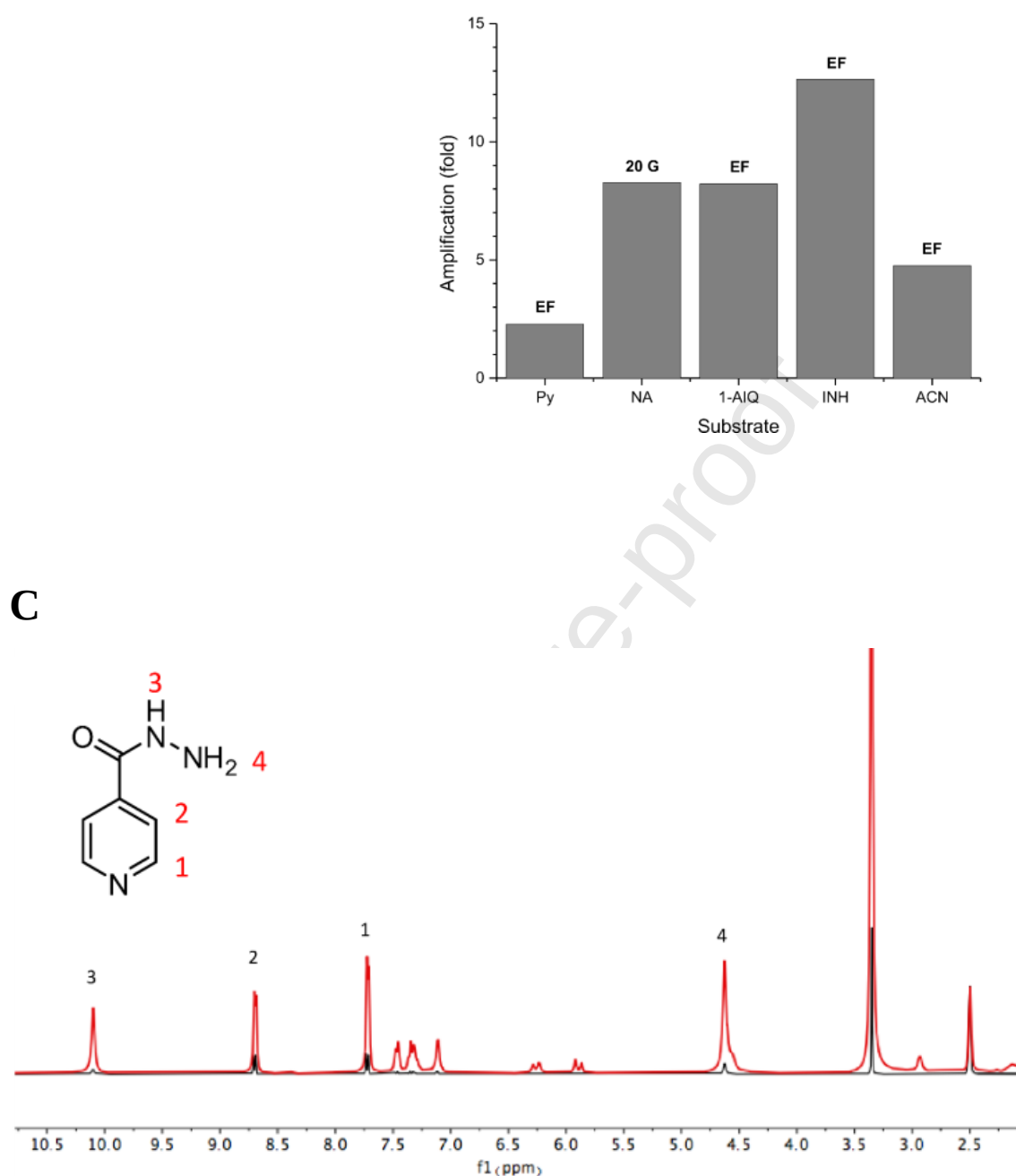


Fig. 3. (a) Structure of the new Ir-Benzimid catalyst. (b) Substrates used for SABRE and the corresponding proton enhancement factors obtained in DMSO at the indicated polarization-transfer fields. “EF” stands for “earth’s field”, which means 0.5G; asterisks mark the most strongly polarized proton used to construct the graph. (c) Thermal-equilibrium (black) and hyperpolarized (red) ¹H NMR spectra of isoniazid (INH) obtained with the Ir-Benzimid catalyst.

3.4. DFT analysis of the catalyst–substrate–parahydrogen complexes

To understand the structural origin of the differing polarization efficiencies, we carried out DFT calculations on the temporary complexes formed between each catalyst, parahydrogen, and the substrates in methanol and DMSO, following approaches established for SABRE-active iridium complexes.^{36–38} The optimized geometries were used as comparative computational models to evaluate relative structural differences among the SABRE complexes under identical computational conditions, rather than as experimental confirmation of the isolated catalyst structures. The resulting average axial and equatorial iridium–substrate bond lengths are summarised in Table 1, and a representative optimized structure is shown in Figure 4. For the Ir-Mes complex, all three bond lengths—the axial bond and the two equatorial bonds—increase on going from methanol to DMSO. Although static bond lengths do not directly determine ligand-exchange kinetics, the longer Ir–substrate bonds may reflect weaker substrate coordination and a greater tendency toward dissociation. If substrate exchange becomes too rapid, the residence time of the bound substrate may become insufficient for efficient polarization transfer, which could contribute to the lower SABRE efficiency of Ir-IMes in DMSO. The Ir-substrate bond lengths calculated for Ir-Imid complexes in DMSO, were, on the whole, longer than of the corresponding Ir-Benzimid complexes. The relatively shorter bonds in the Ir-Benzimid complexes may indicate somewhat stronger substrate coordination and longer residence times, which could place certain catalyst–substrate combinations closer to the optimal exchange regime for SABRE. In addition to these geometric correlations, the imidazol-2-ylidene and benzimidazol-2-ylidene skeletons may exhibit different electronic properties when bound to a metal. Benzimidazolylidene ligands have been reported to be less electron-donating than their imidazolylidene,³⁹ which is consistent with previous findings showing accelerated substrate dissociation in Ir complexes with electron-

donating substituents.⁸ Therefore, substrate dissociation is likely to be relatively slower in Ir-Benzimid complexes than in Ir-Imid complexes. Furthermore, DMSO can directly participate in the coordination sphere of Ir complexes, thereby altering the ligand environment and exchange equilibrium.⁴⁰ Nevertheless, the fact that Ir-Imid exhibited higher efficiency with pyridine and 1-aminoisoquinoline demonstrates that the observed polarization efficiency cannot be fully explained by optimized bond lengths or the electronic properties of the ligands alone. Therefore, it seems that the efficiency of SABRE in dimethyl sulfoxide (DMSO) depends on achieving the right balance between the strength of substrate binding, the rate of ligand exchange, the solvent's ability to coordinate competitively, and the rate of polarization transfer.

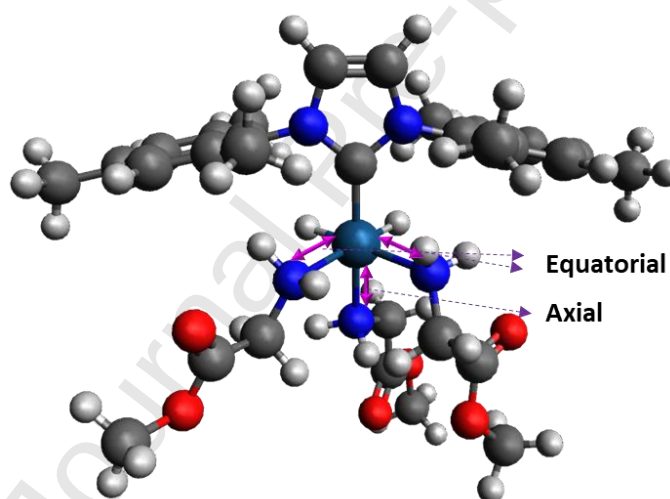


Fig. 4. Representative DFT-optimized structure of a catalyst–substrate–parahydrogen complex, indicating the axial and equatorial Ir–ligand bonds whose average lengths are reported in Table 1.

Table 1. Average axial and equatorial Ir–ligand bond lengths (Å) for the optimized catalyst–substrate–parahydrogen complexes, obtained at the BP86/LanL2DZ level with the SMD solvation model.

Catalyst	Solvent	Axial	Equatorial 1	Equatorial 2
Ir-IMes	MeOH	2.1250	2.1852	2.2344
Ir-IMes	DMSO	2.1253	2.1908	2.2351
Ir-Imid	DMSO	2.1279	2.2100	2.2118

Catalyst	Solvent	Axial	Equatorial 1	Equatorial 2
Ir-Benzimid	DMSO	2.1228	2.2022	2.2108

4. Conclusion

DMSO has recently emerged as an attractive, biocompatible solvent and co-ligand for SABRE, including for relaying polarization to multiple nuclei. The conventional [Ir(IMes)(COD)Cl] catalyst, however, provides only limited enhancement in DMSO, and catalyst development for this medium has been scarce. To address this gap, we synthesized two new iridium NHC catalysts—Ir-Imid and Ir-Benzimid, which produced substantially higher ^1H SABRE enhancements than Ir-IMes, with improvements of up to approximately threefold. DFT calculations revealed differences in the average axial and equatorial iridium–substrate bond lengths of the corresponding SABRE complexes. Although these optimized bond lengths do not directly determine ligand-exchange kinetics, they suggest that differences in substrate binding and residence time may contribute to the observed enhancement trends. The results suggest that efficient SABRE in DMSO requires an appropriate balance among substrate-binding strength, ligand-exchange kinetics, competitive solvent coordination, and polarization-transfer dynamics, rather than simply the shortest possible Ir–substrate bonds. This balance is influenced by the NHC electronic properties and direct DMSO coordination at the Ir atom. Because proton relaxation is rapid in DMSO, we anticipate that further gains can be realized by shortening the transfer time between polarization and detection and by increasing the parahydrogen partial pressure and para-enrichment. More broadly, this work demonstrates that efficient SABRE catalysts can be designed specifically for DMSO and provides practical structural guidance for doing so, thereby widening the scope of parahydrogen-based hyperpolarization for future bio-NMR and MRI applications.

Appendix A. Supporting Information

The Supporting Information contains the ^1H and ^{13}C NMR characterization spectra of the two new iridium catalysts (Figs. S1–S4); the SABRE-hyperpolarized ^1H NMR spectra of each substrate (pyridine, nicotinic acid, 1-aminoisoquinoline, isoniazid, and acetonitrile) obtained with each catalyst in DMSO (Figs. S5–S9); the magnetic-field-dependent enhancement profiles for all three catalysts and all five substrates over the 0–120 G range (Figs. S10–S24); and the complete DFT-optimized geometries, total energies, dipole moments, and polarizabilities of the catalyst–substrate–parahydrogen complexes.

Acknowledgements

This work was supported by the Korea Research Institute for Defense Technology Planning and Advancement (KRIT) grant funded by the Defense Acquisition Program Administration (DAPA) (KRIT-CT-23-031). Institute of Information & Communications Technology Planning & Evaluation (IITP) grant funded by the Korean government (MSIT) (no. RS-2025-25464788).

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Declaration of Competing Interest

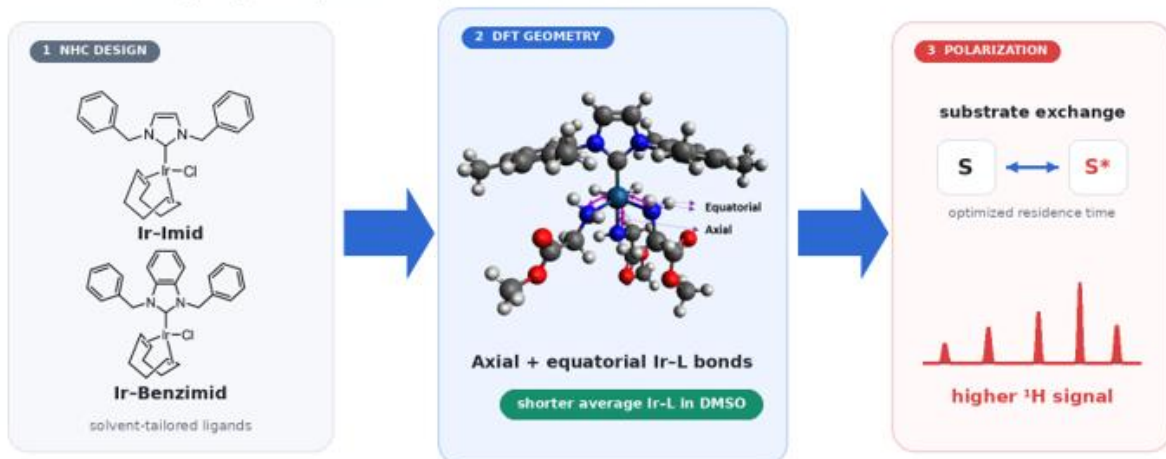
Manuscript title: Efficient Iridium N-Heterocyclic Carbene Catalysts for Signal Amplification by Reversible Exchange in Dimethyl Sulfoxide

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Structure-guided catalyst design

DFT connects Ir-ligand geometry with SABRE performance in DMSO



Highlights

Efficient Iridium N-Heterocyclic Carbene Catalysts for Signal Amplification by Reversible Exchange in Dimethyl Sulfoxide

- Two new Ir-NHC catalysts enable efficient SABRE hyperpolarization in DMSO.
- Ir-Imid and Ir-Benzimid markedly boost ^1H SABRE enhancement in DMSO.
- Enhancements span five substrates, including the TB drug isoniazid.
- Benzimidazole-based catalyst performs well, counter to expectation.
- DFT links efficiency to Ir-ligand bond lengths, guiding catalyst design.